NOTES FOR MATH 7800: TOPICS IN ALGEBRAIC GEOMETRY

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1. Introduction

These notes are intended for advanced graduate students who have allready taken some course in Algebraic Geometry and are familiar with [Hartshorne]. The main topic of this course is the birational classification of higher dimensional complex projective varieties.

Let X be a smooth complex projective variety of dimension n. The main tools that one can use to study X are the tangent bundle T_X, and other vector bundles obtained from this, such as the sheaf of holomorphic i-forms $\Omega^i_X = \Lambda^i T^\vee_X$. Of particular interest is the canonical line bundle $\omega_X = \mathcal{O}_X(K_X) = \Omega^n_X$. One investigates X by considering the pluri-canonical maps

$$\phi_m : X \dashrightarrow \mathbb{P}H^0(\mathcal{O}_X(mK_X)).$$

Some of the natural questions that arise are: what can one say about the topology of these varieties; can the structure of the pluricanonical maps be understood; are there restrictions on the "numbers" that one naturally associates with X; how do these invariants vary in families etc.

When $\dim X = 1$, we say that X is a curve. In this case, the situation is particularly simple. X is topologically classified by its genus g. If $g = 0$ then $X \cong \mathbb{P}^1$. If $g = 1$, then X is an elliptic curve. There is a connected irreducible 1-dimensional family of these. When $g \geq 2$, we say that X is of general type. For each $g \geq 2$ there is a $3g - 3$ dimensional family of these objects. It is not hard to show that if $g \geq 2$, and $m \geq 3$ then the map ϕ_m defines an embedding of X. The numbers $P_m(X) = h^0(mK_X)$ are deformation invariant.

When $\dim X = 2$, we say that X is a surface. One immediately notices that the situation is much more involved. First of all, given a smooth surface X and a point $x \in X$, one can always blow up X at x, obtaining a smooth surface X' and a morphism $p : X' \longrightarrow X$ such that p is an isomorphism over $X - x$ and $\mathbb{P}^1 \cong E \subset X$ is a smooth rational curve with $E^2 = K_X.E = -1$. It is therefore preferable to try and classify smooth surfaces modulo birational isomorphisms. Recall that two surfaces are birational if they contain isomorphic Zariski open subsets. From the topological point of view this is a nontrivial operation, however one sees that varieties in the same birational class share many properties. For example if X, X' are birational, then $h^0(\mathcal{O}(mK_X)) = h^0(\mathcal{O}(mK_{X'}))$ for all $m > 0$. We denote this birational invariant by $P_m(X)$. Other birational invariants include $q(X) = h^0(\Omega^1_X)$ and the Kodaira dimension $\kappa(X) \in \{-\infty, 0, 1, 2\}$ defined as follows If $P_m(X) := h^0(\mathcal{O}_X(mK_X)) = 0$ for all $m > 0$, we set $\kappa(X) = -\infty$. Otherwise let $\kappa(X)$ be the maximum dimension of $\phi_m(X) \subset \mathbb{P}H^0(\mathcal{O}_X(mK_X))$ for all $m > 0$. (In some references the case $\kappa(X) = -\infty$ is refered to as "negative Kodaira dimension" and sometimes by $\kappa(X) = -1$.) It turns out that $\kappa(X) = -\infty$ if and only if X is ruled (i.e. if there is a rational curve through any point of X.) If $\kappa(X) \geq 0$, then there is a surface X' in the birational class of X such that $K = K_{X'}$ is nef, that is $K.C \geq 0$ for any curve $C \subset X'$. Since $\dim X = 2$, there are 3 other possibilities:

(1) $\kappa(X) = 0$ i.e. the maximum of $\{P_m(X) | m > 0\}$ is 1. In fact it turns out that $P_m(X) = 1$ for some $m \in \{1, 2\}$. Then X is birational to an abelian surface ($q = 2$) or a hyperelliptic surface ($q = 1$) or a $K3$ surface ($P_1 = 1, q = 0$) or to an Enriques surfaces ($P_1 = q = 0$).
(2) $\kappa(X) = 1$. In this case X is fibered by elliptic curves, i.e. there is a morphism to a curve $X \longrightarrow B$ such that the general fiber is an elliptic curve.
(3) $\kappa(X) = 2$ and we say that X is of general type. These are essentially impossible to classify as there are infinitely many families. Even if one fixes the invariants q and P_m for all $m > 0$ (so that there are only finitely many families with these given invariants) it is usually hard to determine how many (if any) families of such surfaces exist. Never the less, certain features of these surfaces are well understood. For example ϕ_m is birational as soon as $m \geq 5$.

In higher dimension, the situation is much more complicated. One would of course like to mimic the results described above for surfaces. Ideally, one would like to prove the following:

Conjecture 1.1. *Let X be a smooth projective variety*

(1) *(Mori) $\kappa(X) = -\infty$ if and only if X is uniruled (i.e. covered by rational curves);*

(2) *If* $\kappa(X) \geq 0$*, then* X *has a birational model with mild singularities such that its canonical line bundle* K *is* $\mathbb{Q}$*-Cartier and nef.*
(3) *If* $\kappa(X) \geq 0$*, then the canonical ring* $R(X, K) = \oplus H^0(mK_X)$ *is finitely generated.*

The minimal model program attempts to solve these conjectures. So far it is succesful in answering all of these questions only in dimension at most three. Substantial progress has been made also in higher dimensions and there are many beautiful applications of these techniques to other problems in Algebraic Geometry.